ViT-DR: Vision Transformers in Diabetic Retinopathy Grading Using Fundus Images

N Jagan Mohan
Bio-Medical Imaging Laboratory
Department of Electronics and Communication Engineering
National Institute of Technology Silchar
Silchar, Assam, India
jaganmohan427@gmail.com

R Murugan
Bio-Medical Imaging Laboratory
Department of Electronics and Communication Engineering
National Institute of Technology Silchar
Silchar, Assam, India
murugan.rmn@ece.nits.ac.in

Tripti Goel
Bio-Medical Imaging Laboratory
Department of Electronics and Communication Engineering
National Institute of Technology Silchar
Silchar, Assam, India
triptigoel@ece.nits.ac.in

Parthapratim Roy
Department of Ophthalmology
Silchar Medical College and Hospital
Silchar, Assam, India
parthapratim.smc@gmail.com

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***Abstract*—Diabetic retinopathy (DR) is a complication of diabetes caused by blood leakage into the retinal tissues. DR has become one of the primary causes of visual impairment worldwide, and it is the leading risk factor for poor vision in patients aged 25 to 74. Many symptoms cause retinal degeneration, resulting in vision loss. Early DR detection helps avoid vision loss. Manual DR grading using fundus images is time-consuming and requires expert ophthalmologists. Therefore, we propose Vision Transformer (ViT) based DR severity classification method. In this work, the fundus images are initially divided into non-overlapping patches to retain location information. Then, the flattened patches are converted into sequences before going through a linear and positional embedding process. The generated sequence is then fed into several multi-head attention layers, which produce the final representation. The first token sequence is fed into a softmax classification layer in the classification stage, which produces the recognition output. The proposed work is tested on Kaggle, IDRiD databases. The results obtained are better than the convolutional neural networks and state-of-the-art.**

***Index Terms*—Fundus image, Diabetic Retinopathy, Vision Transformers, Classification**

I. Introduction

Diabetic retinopathy (DR) is a complication of diabetic Mellitus in diabetic patients. According to World Health Organisation, DR is the leading cause of blindness [1]. DR worsens over time and can result in permanent blindness without caution. Diabetes is expected to affect approximately 642 million adults worldwide by 2040, with DR affecting one out of every three diabetics [2]. The study in [3] reports that people with DR will reach 191 million by 2030.

Non-proliferative DR (NPDR) and proliferative DR (PDR) are two main DR stages best described by neovascularization or vitreous/pre-retinal hemorrhage [4] [5]. Each year, 10% of healthy diabetics affect NPDR, while 75% of severe NPDR patients develop PDR. The progression from No DR to PDR takes many years. Further, the NPDR is categorized as mild, moderate, and severe based on the retinal lesions such as Microaneurysms [6], Exudates [7] in the fundus images, as shown in Fig. 1.

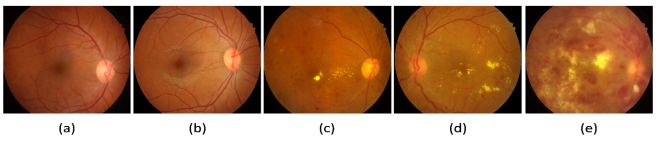


Fig. 1. Typical fundus images representing DR severity (a) Healthy (b) Mild (c) Moderate (d) Sever and (d) PDR

Early DR detection and regular screening help avoid vision loss in diabetic patients. However, manual diagnosing diabetic patients take time for ophthalmologists. The fundus image analysis is frequently used to lower the screening test cost. This allows for computer-aided diagnosis [8] technology, which has been extensively researched in artificial intelligence for healthcare applications [9] [10] [11] [12].

The majority of the efficient features in traditional machine learning (ML) must be identified manually by an expert [11].

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Furthermore, the performance of ML methods depends on the features extracted. Recently, deep learning (DL) algorithms have been widely used in medical image analysis [10], which extracts features automatically. Another hand, the DL techniques are data-hungry require more data to train the model. However, it is difficult in the medical imaging field.

In addition to Convolutional neural networks (CNN) based approaches, a DL method called Transformer is proposed in [13] with good image classification results. The Transformers employ an effective attention mechanism that focuses on specific portions of the input fundus image to achieve more effective performance. Researchers extended the concept of transformers to image classification tasks [14].

Inspired by the fulfillment of Transformers in Natural Language Processing, new studies attempts to use Transformers in image classification tasks. This is a complex task because of the need for self-attention software in that each pixel attends to all different pixels. For images, that is very costly because the images include an extensive wide variety of pixels.

Dosovitskiy et al. [14] applied transformers directly on images and achieved better results with few modifications. Inspired by [14], we propose Vision Transformer (ViT) based DR classification method in this paper. The proposed method is tested using the Kaggle, IDRiD public databases.

The structure of the paper is as follows. A brief review of the DR grading techniques is explained in Section II. Section III describes the database, step-by-step approach to the proposed model. Results and discussion are provided in Section IV. Section V concludes the proposed work with future scope.

II. Literature Review

This section provides a brief overview of the works proposed for DR severity grading. Researchers have proposed several techniques based on standard ML techniques for DR grading. Standard ML techniques employ manual features appropriate for image classification problems with a small database. However, ML techniques are time-consuming and require feature engineering. The ML output depends on the feature set selected. However, the results obtained with one database features often fail in classifying the other database. Therefore, automated feature extraction is performed in deep neural networks such as CNN. Different types of deep architectures were proposed by the researchers in DR classification, including ensemble learning [15] [16], transfer learning [17], hybrid models [18]. The DL techniques help classify the DR severity and perform pixel-level segmentation to detect the retinal lesions.

To the best of our knowledge, very few papers were proposed for DR grading using ViT. Wu et al. [19] proposed ViT based DR grading. In [19], fundus images with 224×224, and $16 \times 16, 32 \times 32$ patch sizes were chosen in grading the DR. However, these models are computationally cost and tested with only one database. AlDahoul et al. [20] proposed encoding fundus data to words using ViT for DR grading. This technique resized the fundus images to 256×256 with 16×16 patches. However, this technique is tested on only one dataset and reported poor performance metrics. Literature reveals that several research gaps exist in grading the DR, such as poor metrics and computational complexity. Therefore, to improve DR grading efficiency, we propose an improved ViT based DR classification model. The proposed model is tested on two public databases such as Kaggle and IDRiD.

TABLE I
DR DATABASE WITH SEVERITY LEVELS

Database	Healthy (Gr-0)	Mild (Gr-1)	Moderate (Gr-2)	Severe (Gr-3)	PDR (Gr-4)
IDRiD	134	20	136	74	49
Kaggle	25770	2432	5278	867	707

III. Methodology

This section presents the database used and the step-by-step approach of the proposed method for DR severity grading.

A. Database

The proposed work is trained and tested with the publicly available databases Kaggle [21], IDRiD [22]. Table I shows the available fundus images in each of the databases. In this work, we have selected 2500 images with each grade consisting of 500 images under the supervision of Ophthalmologists at Silchar Medical College and Hospital.

B. Method

Fig. 2 shows the architecture of the ViTDR. The ViT outperformed in the field of NLP.

In general, the ViT consists of three layers: embedding, encoder, and head classification layer. The standard Transformers proposed in [23] take only 1D token embeddings. To input the fundus images to ViT, firstly, we have reshaped the fundus image (f) $f \epsilon R^{(HXWXC)}$, where H, W represents the resolution of fundus image, C represents the number of color channels into a sequence of flattened 2D patches $f_p \epsilon R^{(N_p \times (P^2.C))}$. Transformers consider each patch as a token. As shown in Fig. 2, in this work, the fundus images are divided into patches and fed to the transformer encoder. Each patch resolution is (P, P) with $N = \frac{HW}{P^2}$ number of patches, serving the Transformer with efficient input sequence length.

1) Linear embedding layer: Prior to feeding the patch sequence into the encoder, the patches are mapped to a constant D dimensional latent vector size passed through the Transformer layers with a trainable linear projection T. The fundus image (f_{class}) label is added to the embedding representation used in DR severity grading. Transformers consider all embedding patches as a set without any order. To preserve the fundus image spatial information of patches, positional data T_{pos} is attached to patches. The embedding

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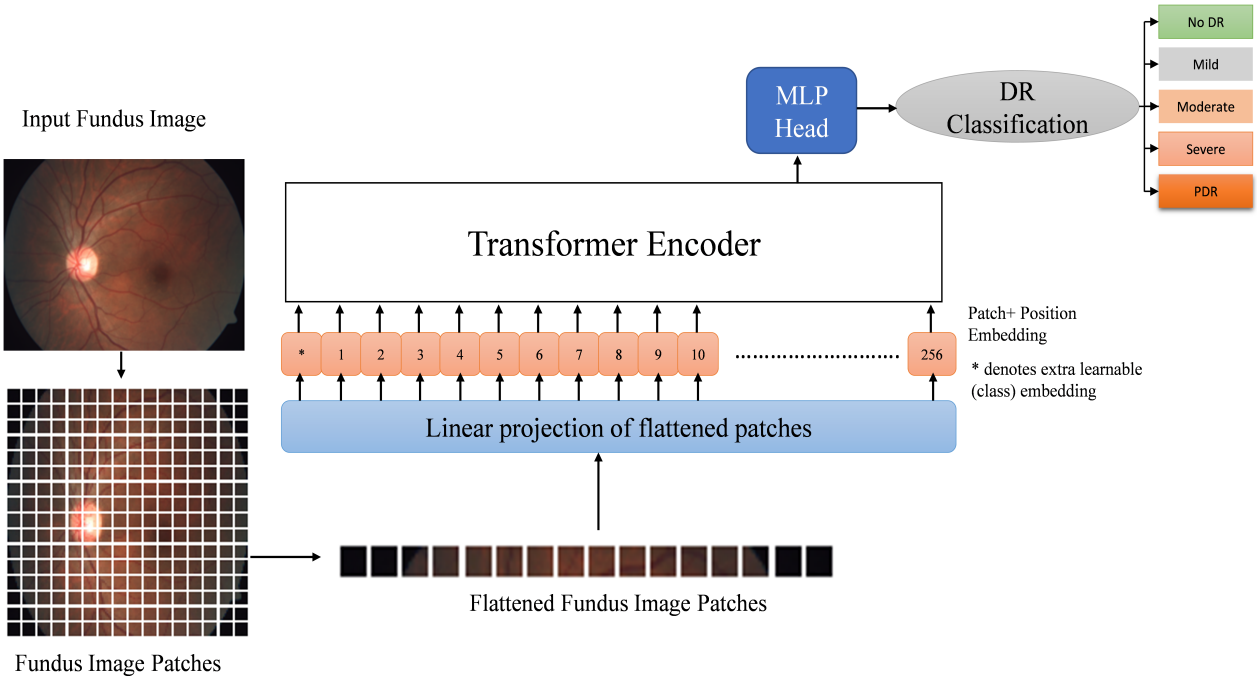


Fig. 2. ViT-DR architecture

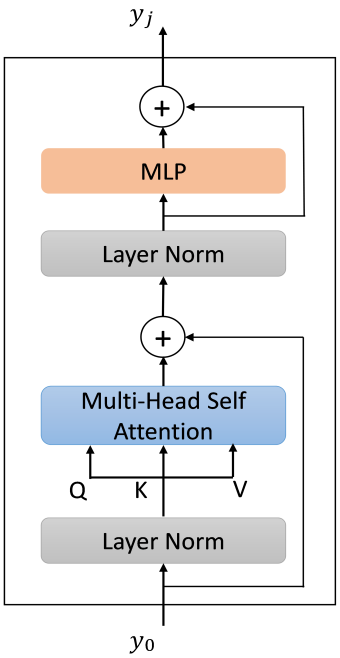


Fig. 3. ViT Encoder

sequence of patches with the token y_0 that resulted is provided in Equation (1).

$$y_0 = [f_{class}; f_p^1 T; f_p^2 T; \ldots\ldots f_p^N T] + T_{pos}, \quad T \epsilon R^{(P^2.C)\times D}, T_{pos} \epsilon R^{(N+1)\times D} \tag{1}$$

The results obtained with 1D and 2D positional encode are same as reported in [14]. Therefore, to preserve flattened patches, positional information 1D encoding is used.

2) ViT encoder: The embedding sequence of patches with the token y_0 is inputted to the encoder. The encoder contains an alternative layer arrangement with multi-headed self-attention (MSA) (represented in Equation (2)), multi-layer perceptron (MLP) (represented in Equation (3)), and layer norm (LN). The final block consists of a Gaussian error Linear Unit (GeLU) layer placed between two dense layers. As shown in Fig. 3, the encoder contains the residual skip connections followed by LN. The encoder comprises of J identical layers.

$$y_j^{'} = MSA(LN(y_{(j-1)})) + y_{(j-1)}, j = 1\ldots..J \tag{2}$$

$$y_j = MLP(LN(y_j^{'})) + y_j^{'}, j = 1\ldots..J \tag{3}$$

For predicting the DR severity grading class, we consider the sequence y_J^0 at encoder last layer and pass it to the classifier.

$$Y = LN(y_J^0) \tag{4}$$

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The MSA block determines the relative relevance of a single patch embedding compared to the other embeddings in the sequence. The MSA block consists of linear, self-attention (SA), and the concatenation layer.

The SA head calculates attention weight by computing the weighted sum of all values over the sequence y by finding the query (Q), key (K), value (V) scale dot product. Q, K, and V are generated with learned matrices U_{QKV} using Equation (5) for each sequence inputted. As depicted in Equation (6), the scaled dot product results are fed into softmax with a scaling factor S_K. Finally, Equation (7) is used to locate the patch with the highest attention score.

$$[Q, K, V] = yU_{QKV}, U_{QKV} \epsilon R^{(D \times 3S_K)} \tag{5}$$

$$X = softmax(\frac{(QK^T)}{\sqrt{S}_K}), X \epsilon R^{(n \times n)} \tag{6}$$

$$SA(y) = X.V \tag{7}$$

C. Experimental setup

The original fundus images are resized (128×128) and augmented with random flip, rotation, and zoom. The patch sizes are generally selected as $16 \times 16 \quad or \quad 32 \times 32$ in ViT. However, in the present work, we have selected a patch size of (P, P) with P=8. As the patch size decreases, the sequence length increases. Therefore, this work overcomes the problem of longer sequences by resizing the fundus images to 128×128, with 256 patches per image and 192 elements per patch. This work uses 8 transformer layers with a learning rate of 0.001, batch size of 32, and projection dimension of 64 with 4 MLP heads. AdamW optimizer is used with a weight decay of 0.0001.

IV. Results and Discussion

This section presents the performance metrics, results obtained, and comparison of the proposed model with pre-trained and other models.

A. Performance metrics

The efficiency of the proposed work is measured in terms of Accuracy (Acc) [24], Recall, and F1-score [8]. The metrics definition is depicted using Equations (8)-(10).

$$Acc = \frac{TrPos + TrNeg}{TrPos + TrNeg + FlsPos + FlsNeg} \tag{8}$$

$$Recall = \frac{TrPos}{TrPos + FlsNeg} \tag{9}$$

$$F1 = \frac{2TrPos}{2TrPos + FlsPos + FlsNeg} \tag{10}$$

Where $TrPos$ accurately identifies the presence of DR grade, whereas $TrNeg$ identifies the absence of DR grade. $FlsPos$ incorrectly implies the presence of DR grade, whereas $FlsNeg$ suggests the absence of DR grade.

TABLE II
Severity-wise DR classification results with proposed method

DR Grade	Acc (%)	Recall	F1-score
No DR (Healthy Gr-0)	97.33	88	0.93
Mild (Gr-1)	93	85	0.83
Moderate (Gr-2)	94	83	0.85
Sever (Gr-3)	94.33	88	0.86
PDR (Gr-4)	96	92	0.90

TABLE III
Proposed model vs. Pre-trained networks performance

Method	Acc (%)	Recall (%)	F1-score
AlexNet	74.2	74.33	0.74
Vgg19	77.33	77.2	77.4
ResNet101	85.67	85.6	0.858
MobileNetV2	81.3	81	0.812
InceptionV3	84.4	84.1	0.84
Proposed model	87.41	87.2	0.874

B. Results

This section presents the results obtained with the proposed ViT model in grading the DR. In this work, the fundus images are initially divided into fixed-size patches and then flattened. Then, the position embedding is added to each patch after creating a low dimensional linear embedding. These patches are then fed to the encoder. The AdamW is used to fine-tune the model to achieve better DR classification results. The proposed model has achieved Acc of 87.41%, Recall of 87.2%, F1-score of 0.874 with augmented fundus data, Acc of 84%, Recall of 83.8%, F1-score of 0.84 with original fundus data. Fig. 4 shows the Acc obtained for each epoch. The DR severity-wise classification results of the proposed model are depicted in Table II. Table II shows that the Gr-0 has achieved the highest classification Acc of 97.33% with a Recall of 88% and F1-score of 0.93. The Gr-1 has a low classification Acc of 93% with a Recall of 85% and an F1-score of 0.83.

An ensemble of ViT model proposed in [20] reported Recall of 45%, F1-score of 0.42, and precision of 47% with a low inference time of 1.12 sec. The work proposed in [19] reported an Acc of 91.4%. However, this model requires an average computational time of more than 9 hours. The methods proposed in [20] [19] were tested only on a single database. Compared to [20] the proposed model reported better performance metrics. The total time to train and test the proposed model is 3.5 hours which is better than [19] with comparable performance metrics.

C. Comparison of the proposed model with pre-trained networks

This section compares the efficiency of the proposed model with the pre-trained CNN models. Table III represents the results of the pre-trained models and proposed model.

Table III reveals that the proposed model has achieved an Acc of 87.41% compared to other networks. The ResNet101

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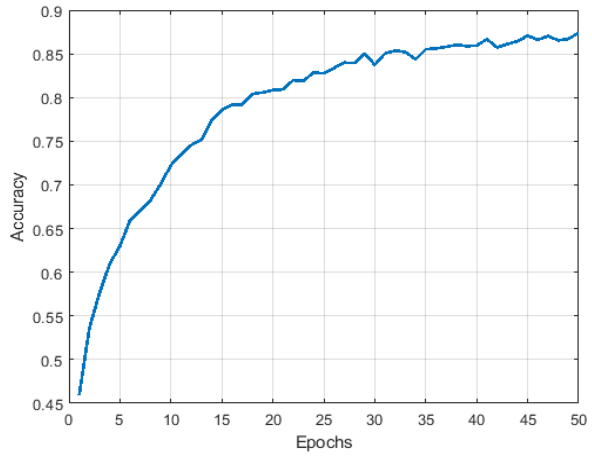


Fig. 4. Proposed model Acc vs. Epochs curve

and InceptionV3 networks have shown similar results to the proposed model. From the experiments, it is observed that as the database increases, the performance of the ViT model improves. However, the computational time and resources required also increase. Therefore, we have used augmented data to increase the database with optimal requirements in the present work. Fig. 5 shows the Reciever operating characteristics curve of the proposed model with pre-trained networks.

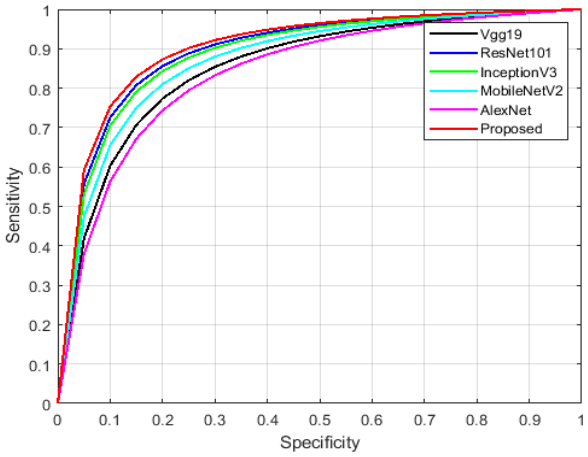


Fig. 5. ROC curve for proposed model vs. Pre-trained networks

D. Discussion

The results reveal that the proposed model helps in grading the DR. The performance of the proposed ViT model is improved with the increasing database using data augmentation. The Acc obtained with the original and augmented databases using the proposed model is 84%, 87.41% presented in the Results section. From Fig. 4, it is observed that the performance of the proposed model can be increased with an increased database and the proper selection of training parameters. Table II depicts the severity-wise DR classification results. The results show that Gr-0 has obtained a high Acc of 97.33%, Gr-1 has obtained a low Acc of 93%. A comparative analysis of the proposed model with pre-trained networks is presented in Table III. Table III reveals that the proposed model achieved better results than pre-trained models with the increased database. It is also observed that the fundus images with greater size with small patches provide superior results. The proposed method is evaluated on two datasets and outperformed existing state-of-the-art techniques.

V. Conclusion and Future Works

Diabetic retinopathy causes irreversible visual loss if not detected and treated on time. This paper presents an improved transformer model for grading the severity levels of diabetic retinopathy. The performance of the proposed model is improved by selecting the proper fundus image size, patch size, number of transformers, multi-layer perceptron head layers. After performing several experiments, we have considered the optimal values for training the model, which gave better results with less computational cost. The proposed model is tested on two public databases. The results achieved with the proposed model are better than the other approaches.

The performance of the proposed model can be improved further by increasing the size of the database and patches at the cost of computation. Preprocessing the fundus images to identify microaneurysms helps increase the diabetic retinopathy classification results. In the future, the efficiency of the proposed model can be improved with better preprocessing techniques and can be tested with a real-time database.

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